

SCIF2001: Research Roadmap Brief

Summary of Information

Assessment title: Research Roadmap

Weighting: Must achieve an overall mark of Competent on this assignment to pass the course.

Due Date: 9pm Friday of Week 4 (9 October 2026)

Group work: Everyone must submit their own unique assignment. You are welcome to discuss methodology and structure of the assignment with others but any work you submit must be your own.

Length: No more than 2,000 words, including figure captions. References are not included in the word count. Any words beyond 2,000 will not be considered in the marking process.

Submission requirements: Single PDF uploaded to Moodle

Feedback Details: Feedback will be provided through a 15-minute one-on-one meeting with your mentor, typically within approximately ten working days of submission. Meetings will be recorded, with transcripts made available to support your processing of verbal feedback. Students are expected to engage actively in these feedback sessions and take responsibility for summarising and reflecting on feedback received. Guidance on maximising the value of verbal feedback will be provided through dedicated resources in the course modules.

Aligned CLOs: This assignment is related to CLOs 2, 3, and 4. In this assignment, you will “Refine an individual research question and approach, considering the available research tools and relevant scientific literature” and “Communicate approaches, outcomes and implications of a scientific investigation.” Between this submission and your feedback meeting with your Mentor, you will “Respectfully discuss relevant scientific topics through a variety of formats.”

Assessment Description

Rationale

Research is never performed in a vacuum, it builds on previous work and answers some question that has been posed as a result of other studies, theories, or analyses.

The Research Roadmap is your chance to:

- Turn a broad project idea into a focused, answerable research question,
- Explore relevant literature,
- Design a practical methodology, and
- Map out a realistic timeline.

Just like professional scientists, you’ll learn to define a problem, situate it in existing research, and chart a feasible path forward. All this in no more than 2,000 words!

Task description

You will submit a structured research proposal for your project via Moodle. The proposal will help you develop a clear and well-supported research strategy, including a research question, literature review, methodology, and project plan. In preparing your proposal, you are expected to identify and discuss relevant scientific issues, such as differing approaches, interpretations, or perspectives within your research area. You will have the opportunity to seek support from your mentor and peers in preparing your individual submission.

This proposal will have four parts:

1. Research Question – a clear, specific, and feasible question you aim to answer, with an explanation of why it is important.
2. Literature Review – key background, critical references, and why this question matters.

3. Methodology – the approach and methods you will use, including potential challenges.
4. Project Plan – a week-by-week outline for Weeks 5–10: milestones, roles, and risk management.

Figures or diagrams are strongly encouraged: flow charts, schematics, or visuals often communicate complex plans better than words.

Tips for Success

- Start early. Dig into background papers early so you can focus on writing in Week 4.
- Think like a scientist. Why is this question important? How will your approach tell us something new?
- Be concise and clear. Keep your eyes on the prize, your plans focused, and your timeline realistic.
- Show, don't just tell. Use visuals to make your plan easy to follow.
- Collaborate, then individualise. Discuss methods with your team, but write your own proposal in your own words.

Submission details

You will submit all materials into a portal on Moodle before the deadline.

Resources

See Moodle for examples of proposals from across Science that might help you understand how we frame research questions and our attempts to answer them. Ask questions to your Mentors and Leads, as well as to your project team. Note the rubric below and make sure you are addressing all the points on it.

Marking rubric

This assessment uses a competency-based framework. Each criterion will be assessed as Competent with Merit (CM), Competent (CP), or Not Yet Competent (NC).

Competent (CP) represents the expected standard for successful completion of the assessment. Competent with Merit (CM) recognises work that goes meaningfully beyond this standard through stronger scientific judgement, critical synthesis, justification, or anticipation of challenges. Not Yet Competent (NC) indicates that further development is required to demonstrate the relevant capability.

The first four criteria are core competencies because together they establish whether the proposed research is scientifically justified, feasible, and ready to proceed. Structure & Presentation and Use of Figures & References are supporting competencies that contribute to the effectiveness and professionalism of the proposal.

Criterion	Competent with Merit (CM)	Competent (CP)	Not Yet Competent (NC)
CORE: Research Question	Presents a precise, focused and feasible research question that demonstrates strong insight into the project context. The rationale clearly establishes the scientific significance of the question and shows thoughtful consideration of what can realistically be answered within the project.	Presents a specific, focused and answerable research question that is well matched to the project scope and available resources. The rationale uses relevant scientific context to clearly explain why the question is worth answering and what the investigation is intended to establish.	The research question is unclear, overly broad, insufficiently connected to the project, or not realistically answerable within the available time and resources. The rationale does not yet establish the scientific purpose or significance of the investigation.
CORE: Literature Review	Critically synthesises relevant, high-quality scientific literature to establish the current state of knowledge, identify an important gap, uncertainty or unresolved issue, and clearly position the proposed research within that context.	Uses an appropriate range of relevant scientific literature to establish the background and motivation for the project. The review demonstrates understanding of the principal findings, approaches or debates in the area, makes meaningful connections between sources rather than simply describing them individually, and uses the literature to justify the research question.	The literature considered is too limited, poorly selected or primarily descriptive. Connections between sources and the proposed research are weak, or important scientific context required to justify the research question is missing or misunderstood.
CORE: Methodology	Develops a coherent and well-justified methodology that clearly connects the proposed methods to the research question. Demonstrates strong understanding of the tools or techniques involved and thoughtfully anticipates limitations, uncertainties, potential challenges and, where appropriate, alternative approaches.	Develops a clear, logically sequenced and feasible methodology capable of answering the research question. The choice of major methods, tools or techniques is scientifically justified rather than simply stated. Important assumptions, limitations and practical challenges are identified, together with reasonable strategies for addressing them where possible.	The proposed methodology is incomplete, unclear, insufficiently justified, or unlikely to answer the research question. Important methodological steps, assumptions, limitations or practical constraints have not yet been adequately considered.
CORE: Project Plan & Feasibility	Presents a realistic and well-integrated week-by-week plan with clear milestones and responsibilities. Dependencies, risks and contingency strategies are thoughtfully considered, demonstrating strong judgement about how the project can be completed within the available time.	Presents a credible and sufficiently detailed week-by-week pathway to completion, with clear milestones, responsibilities and appropriate sequencing of tasks. Important dependencies and risks are identified, and the plan includes realistic responses or contingencies for significant foreseeable problems.	The plan does not yet provide a credible pathway to completing the project. Important milestones, responsibilities, dependencies or risks are unclear or missing, the sequencing is unrealistic, or the proposed scope is inconsistent with the available timeframe.

Criterion	Competent with Merit (CM)	Competent (CP)	Not Yet Competent (NC)
SUPPORTING: Structure & Presentation	Communicates the proposed research exceptionally clearly, concisely and effectively. The document has a coherent scientific narrative in which the research question, literature, methodology and project plan reinforce one another. Organisation and presentation choices consistently support understanding.	Communicates the proposal clearly, concisely and professionally for a scientific audience. The document has a logical overall argument, sections are meaningfully connected, and the reader can readily understand how the literature motivates the question, how the methods address it, and how the plan will deliver the proposed research.	The organisation, writing or presentation makes important aspects of the proposed research difficult to understand. Connections between major components are unclear, or communication problems materially interfere with evaluation of the scientific plan.
SUPPORTING: Use of Figures & References	Selects and integrates references and, where useful, figures or diagrams strategically to strengthen the scientific argument and improve understanding of the research plan. Sources are authoritative and used accurately, and scholarly referencing is clear and consistent.	Uses appropriate and credible scientific sources throughout the proposal to support substantive claims and decisions. References are accurately represented, consistently acknowledged and integrated into the argument. Figures or diagrams, where useful, are clear, purposeful and explicitly connected to the discussion.	The evidence base is insufficient, poorly selected or inaccurately represented; important claims are unsupported; or referencing does not meet the expected scholarly standard. Figures or diagrams, where needed or used, do not effectively contribute to understanding.

Determining the overall outcome

Competent with Merit (CM): All four core competencies are at least Competent (CP), at least three of the four core competencies are Competent with Merit (CM), and both supporting competencies are at least Competent (CP).

Competent (CP): All four core competencies demonstrate the Competent (CP) standard. The proposal, taken as a whole, provides a scientifically justified and feasible basis for proceeding with the research. Supporting competencies must be sufficient for the scientific reasoning and research plan to be clearly evaluated.

Not Yet Competent (NC): One or more core competencies have not yet reached the required standard, meaning that an aspect of the research question, evidence base, methodology or project plan requires substantive further development before the proposed research is ready to proceed. Significant deficiencies in supporting competencies may also result in Not Yet Competent (NC) where they prevent the scientific work from being adequately communicated or evaluated.

Where work sits between two descriptors, the assessor will determine which descriptor best represents the demonstrated level of capability as a whole, rather than awarding points for individual features.

Length requirement

The 2,000-word maximum is a submission requirement rather than a competency criterion. Figure captions are included in the word count and references are excluded. In accordance with the assessment instructions, words beyond the 2,000-word maximum will not be considered during assessment.